AI-Based Headache Pattern Analysis & Migraine Type Prediction

A machine learning decision-support system for headache classification and migraine likelihood assessment

ARTIFICIAL INTELLIGENCE | HEALTHCARE ML

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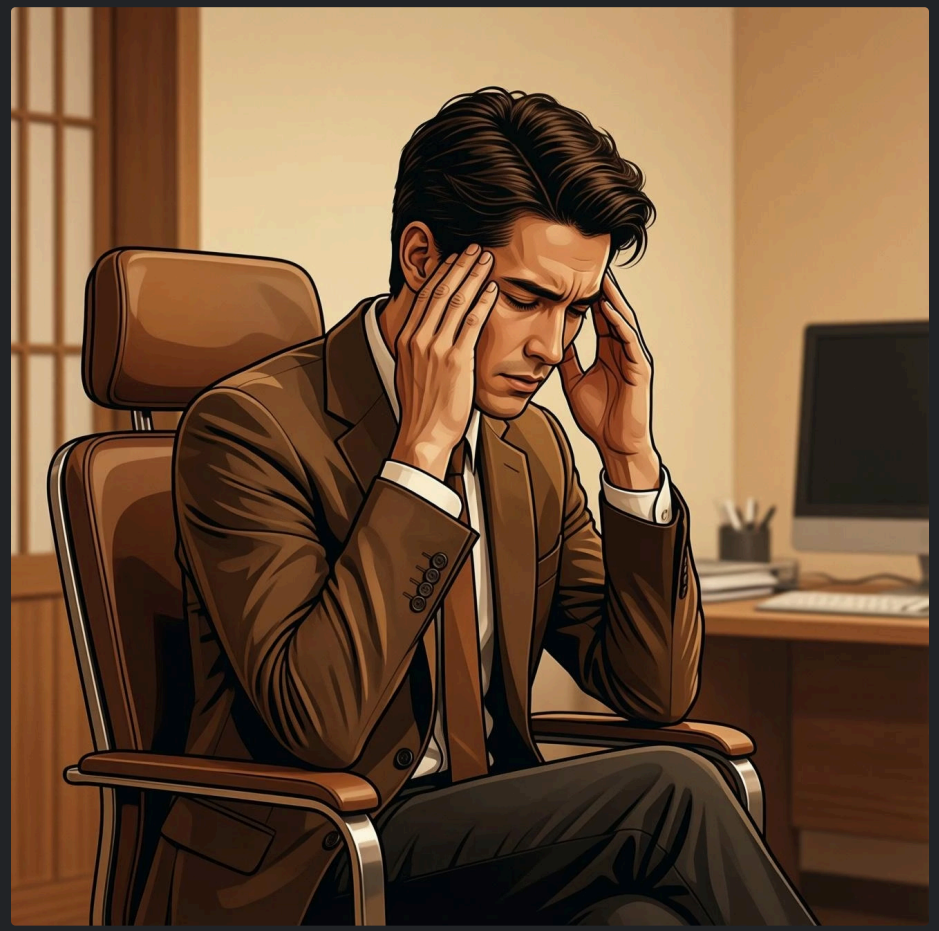
CLINICAL CHALLENGE

The Growing Challenge of Headache Disorders



Headache and migraine disorders represent a significant global health burden, affecting millions and substantially reducing quality of life and workplace productivity.

The complexity lies in **overlapping symptom profiles** across different headache types, making accurate classification challenging even for experienced clinicians.

Misclassification can lead to inappropriate treatment strategies and prolonged patient suffering.

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PROJECT OBJECTIVES

Building an Intelligent Decision-Support System

Core Objective

Develop an AI-powered system to analyse headache symptoms, behavioural patterns, and environmental triggers for accurate classification

Predictive Capability

Calculate probability-based predictions for migraine likelihood and specific headache categories using machine learning

Clinical Support

Provide clinicians with data-driven insights whilst maintaining clear boundaries as a support tool, not a diagnostic replacement

Important: This system is designed to support, not replace, professional medical diagnosis and clinical judgement.

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TECHNICAL APPROACH

Machine Learning Methodology

Data Pipeline

01

Data Collection

Gather symptom profiles, triggers, and patient history

02

Preprocessing

Clean, normalise, and engineer relevant features

03

Model Training

Apply supervised learning algorithms

04

Validation

Evaluate performance using clinical metrics

ML Models Employed

- **Random Forest:** Ensemble method for robust classification
- **Support Vector Machine:** High-dimensional pattern recognition
- **Gradient Boosting:** Sequential error correction for accuracy

Key Input Features

Symptom severity, duration, frequency, known triggers, temporal patterns, and associated symptoms

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SYSTEM ARCHITECTURE

From Symptoms to Predictions

The system processes multiple symptom dimensions through our ensemble of machine learning models to generate probability distributions across headache categories, providing clinicians with quantitative decision support.

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CLASSIFICATION RESULTS

Headache Category Predictions

The system classifies headaches into four primary categories, each with distinct clinical characteristics and treatment approaches:

1

Migraine with Aura

Visual or sensory disturbances preceding headache onset

2

Migraine without Aura

Unilateral throbbing pain without preceding symptoms

3

Tension-Type Headache

Bilateral pressure or tightening sensation

4

Cluster Headache

Severe unilateral pain with autonomic symptoms

Model Performance

Evaluation metrics demonstrate clinical viability:

- **Accuracy:** Overall correct classification rate
- **Precision:** Positive prediction reliability
- **Recall:** True positive detection rate
- **F1-Score:** Harmonic mean of precision and recall

Cross-validation ensures model generalisation across diverse patient populations.

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APPLICATIONS & ETHICS

Clinical Applications & Ethical Framework

Patient Self-Monitoring

Enable individuals to track patterns and recognise early warning signs for proactive management

Clinical Decision Support

Provide clinicians with data-driven insights to complement diagnostic expertise

Ethical Boundaries

Designed explicitly as an assistive tool that augments, rather than replaces, medical judgement

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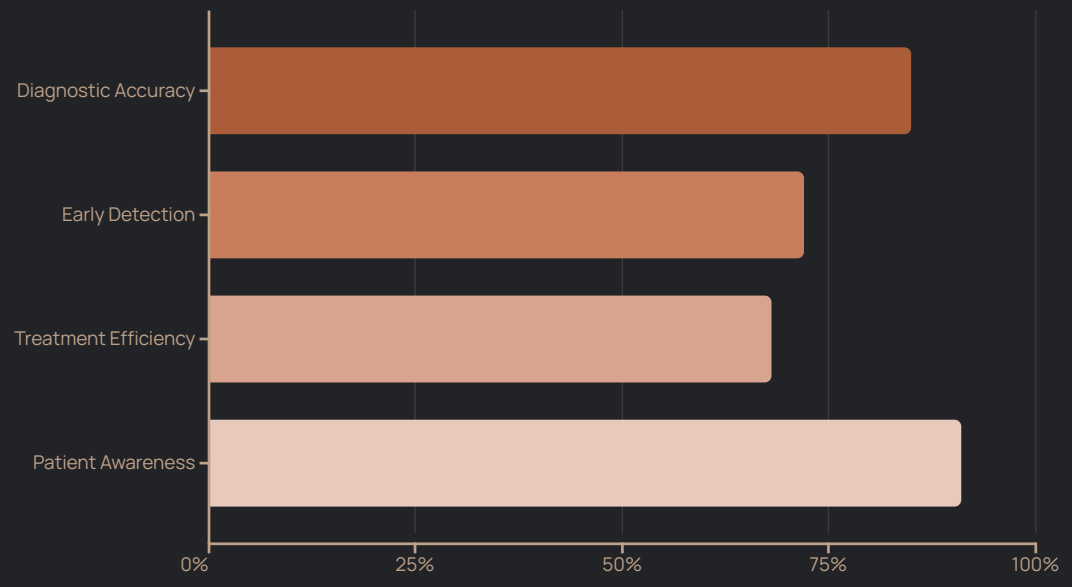
IMPACT ASSESSMENT

Quantifying Clinical Value



Key Benefits

The system demonstrates substantial potential for improving headache management across multiple dimensions.

Enhanced awareness: Patients gain deeper understanding of their symptom patterns and potential triggers.

Informed consultations: Clinicians receive structured, quantitative data to complement clinical assessment.

Resource optimisation: More efficient triage and treatment pathway selection reduces healthcare burden.

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FUTURE DEVELOPMENT

Research Roadmap & Future Enhancements

1 Explainable AI

Implement interpretable models to provide transparent reasoning for predictions, enhancing clinical trust

2 Temporal Analysis

Develop time-series tracking capabilities to identify long-term patterns and seasonal variations

3 Mobile Integration

Create patient-facing mobile application for real-time symptom logging and pattern visualisation

4 Multimodal Data

Incorporate physiological sensors, environmental data, and genetic markers for personalised prediction

Future iterations will focus on enhancing model interpretability, expanding data sources, and improving accessibility through mobile platforms whilst maintaining rigorous clinical validation standards.

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Conclusions & Key Takeaways

This project demonstrates the viability of machine learning approaches for headache pattern analysis and migraine prediction, offering a valuable complement to traditional clinical assessment.

Core Achievements

- Developed ensemble ML system achieving clinically relevant classification accuracy
- Established ethical framework prioritising clinical support over replacement
- Identified clear pathways for real-world deployment and future enhancement

Impact Summary

4

Headache Categories

Classified with probability-based predictions

3

ML Algorithms

Ensemble approach for robust performance

Clinical Significance: Enhanced diagnostic support, improved patient awareness, and optimised treatment pathways

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Thank you for your attention. Questions and feedback welcome.

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